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Multimodal Integration for Heart Transplant Rejection

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Land acknowledgement

I would like to acknowledge that I work on the traditional, ancestral, and unceded territory o the Coast Salish Peoples, including the territories of the xwməθkwəy̓əm (Musqueam), Skwxwú7mesh (Squamish), Stó:lō and Səl̓ílwətaʔ/Selilwitulh (Tsleil-Waututh) Nations.

Traditional: Traditionally used and/or occupied by Musqueam people

Ancestral: Recognizes land that is handed down from generation to generation

Unceded: Refers to land that was not turned over to the Crown (government) by a treaty or other agreement

Background

Heart Transplant

- Replace person’s sick heart with healthy one from someone who has died
- Happens only if heart failure in end-stage and no longer treatable in any other way
- Occurs across all ages, including children
- May lead to immune-mediated rejection of the donor heart (HTR)

Adopted from https://www.providencehealthcare.org/en/health-services/services/surgery-procedures-treatment/heart-transplant

Heart Transplant Rejection Outcomes

- Acute Cellular Rejection (ACR) – T cells attack heart tissue (acute)
- Antibody-Mediated Rejection (AMR) – Antibodies attack blood vessels (acute or recurrent)
- Cardiac Allograft Vasculopathy (CAV) – Chronic narrowing / scarring of vessels from repeated immune attacks
- Quilty Effect – Localized lymphocyte clusters, usually benign

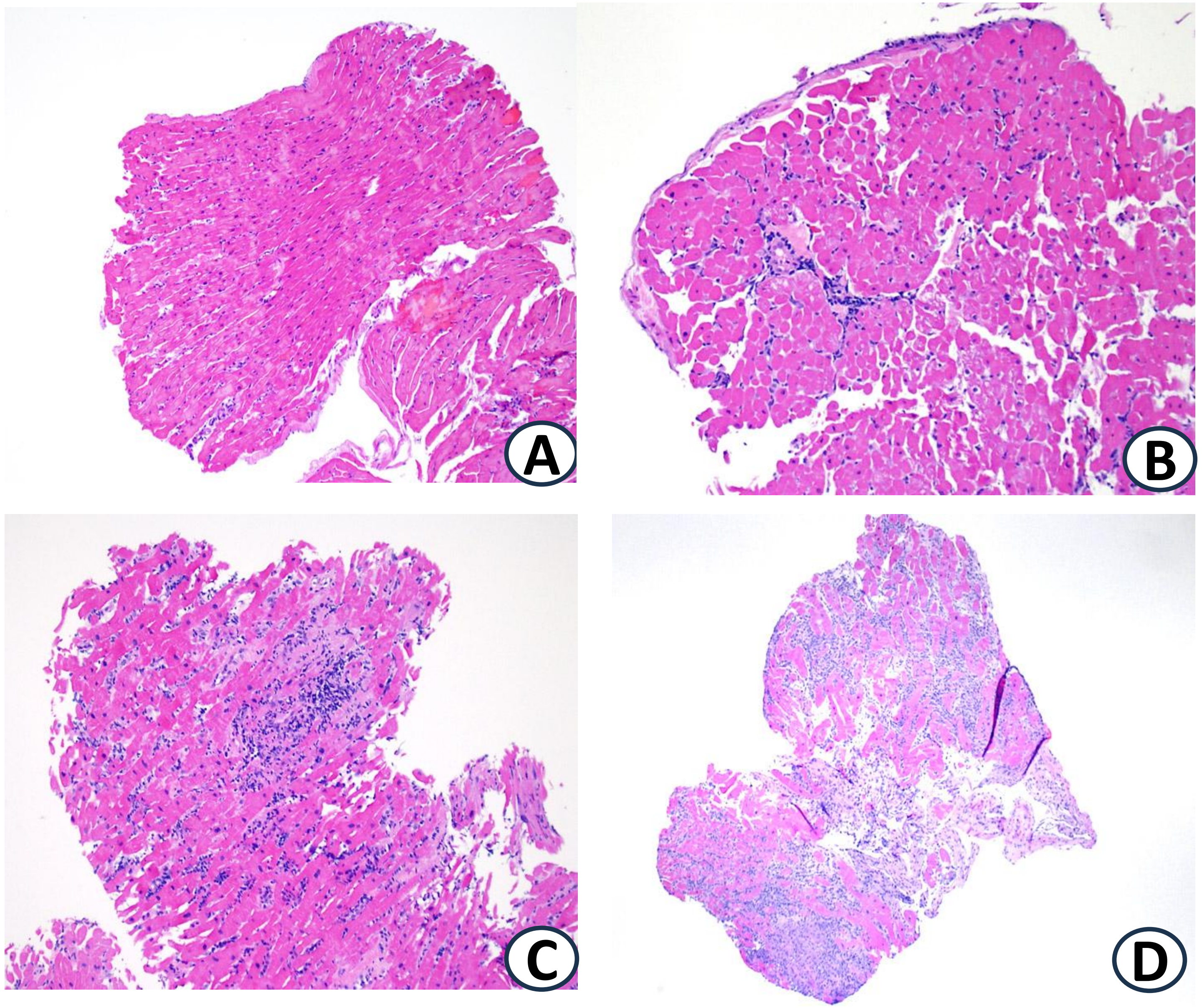
Type	Analogy	Timing
ACR	Soldiers storming heart tissue	Early, acute
AMR	Guided missiles attacking vessels	Early / recurrent
CAV	Scarred battlefield	Long-term, chronic
Quilty	Loitering soldiers in corner	Early, benign

Levels of ACR

Table 1. ISHLT Standardized Cardiac Biopsy Grading: Acute Cellular

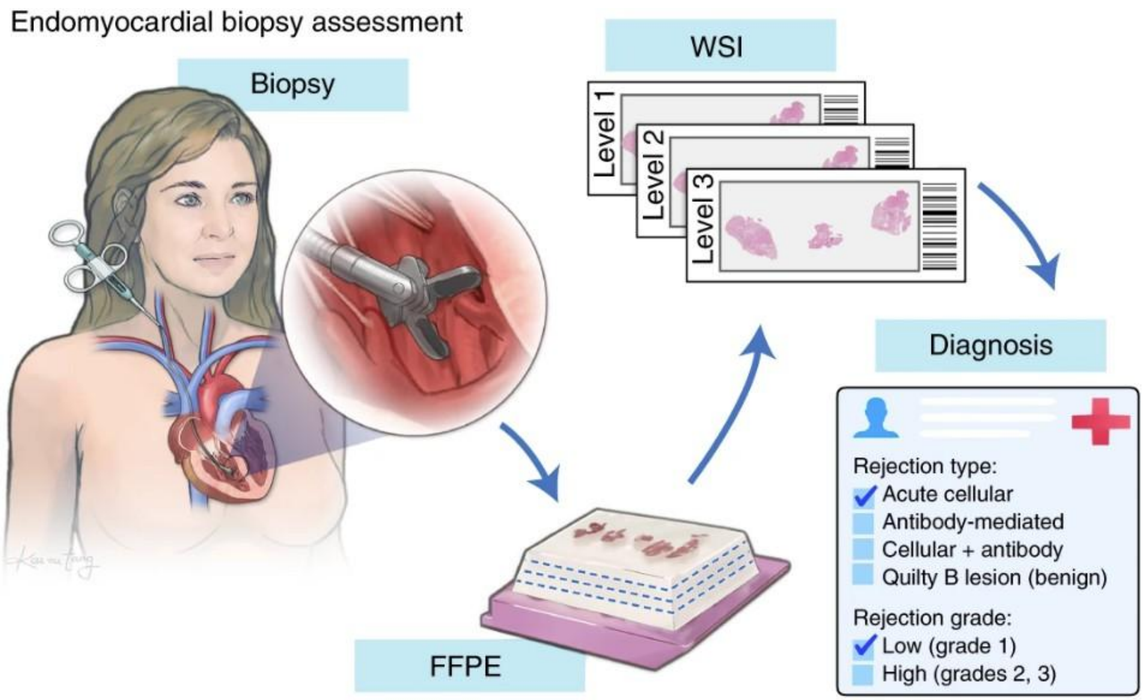
	2004	
A	Grade 0 R[a]	No rejection
B	Grade 1 R, mild	Interstitial and/or perivascular infiltrate with up to 1 focus of myocyte damage
C	Grade 2 R, moderate	Two or more foci of infiltrate with associated myocyte damage
D	Grade 3 R, severe	Diffuse infiltrate with multifocal myocyte damage ± edema, ± hemorrhage ± vasculitis

Stewart et al. 2005 – ACR diagnostic criteria (1)



Adopted from https://www.scvp.net/acute-cellular-rejection-tutorial/#grading-criteria

HTR Diagnosis Challenge

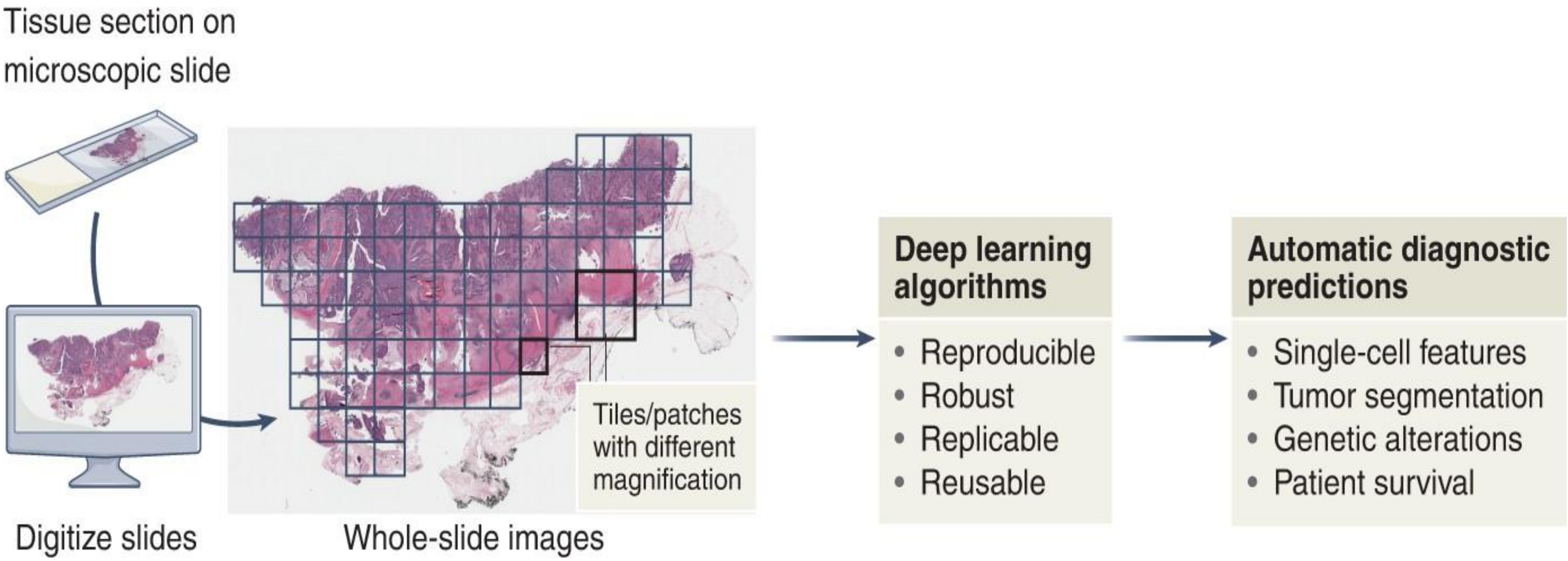


Lipkova et al. 2022 (4)

Kappa (κ)	Interpretation (Landis & Koch, 1977) (5)
< 0.00	Poor agreement
0.00–0.20	Slight agreement
0.21–0.40	Fair agreement
0.41–0.60	Moderate agreement
0.61–0.80	Substantial agreement
0.81–1.00	Almost perfect agreement

- Relies on endomyocardial biopsy (EMB)
- Assessments from pathologists, subjective and opinion-based
- High Interobserver variability with standardized grading (6) :
  - 1990 ISHLT grading system with κ = 0.31
  - 2005 ISHLT grading system with κ = 0.39

Digital pathology / AI on H&E (whole-slide imaging)



Wagner et al. Workflow in computational pathology (7)

Hypotheses & Objectives

We hypothesize that a multimodal approach combining digital pathology with clinical data, laboratory tests and blood biomarkers can:

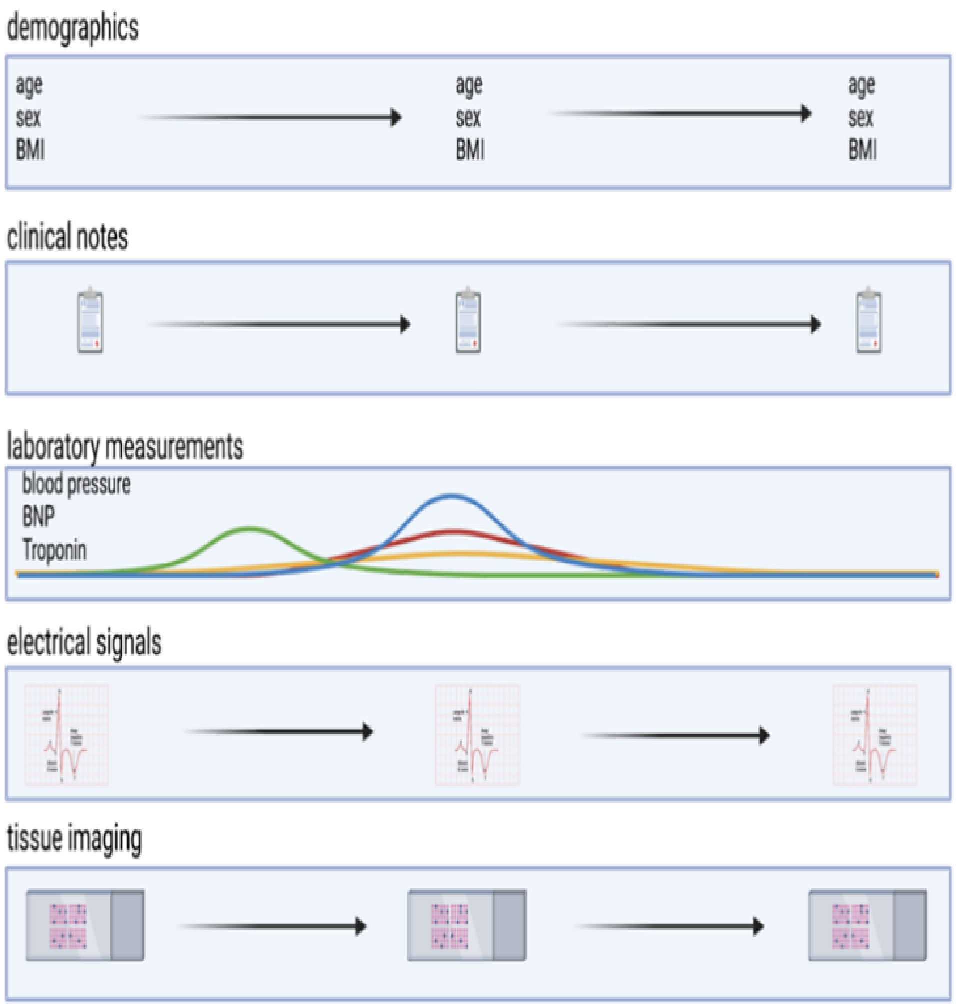
1) Improve heart transplant rejection diagnosis

2) Identify patient trajectories with distinct long-term clinical outcomes

Methods -- Cohort

2005
First patient data collected

2025
Ongoing digitization of slides

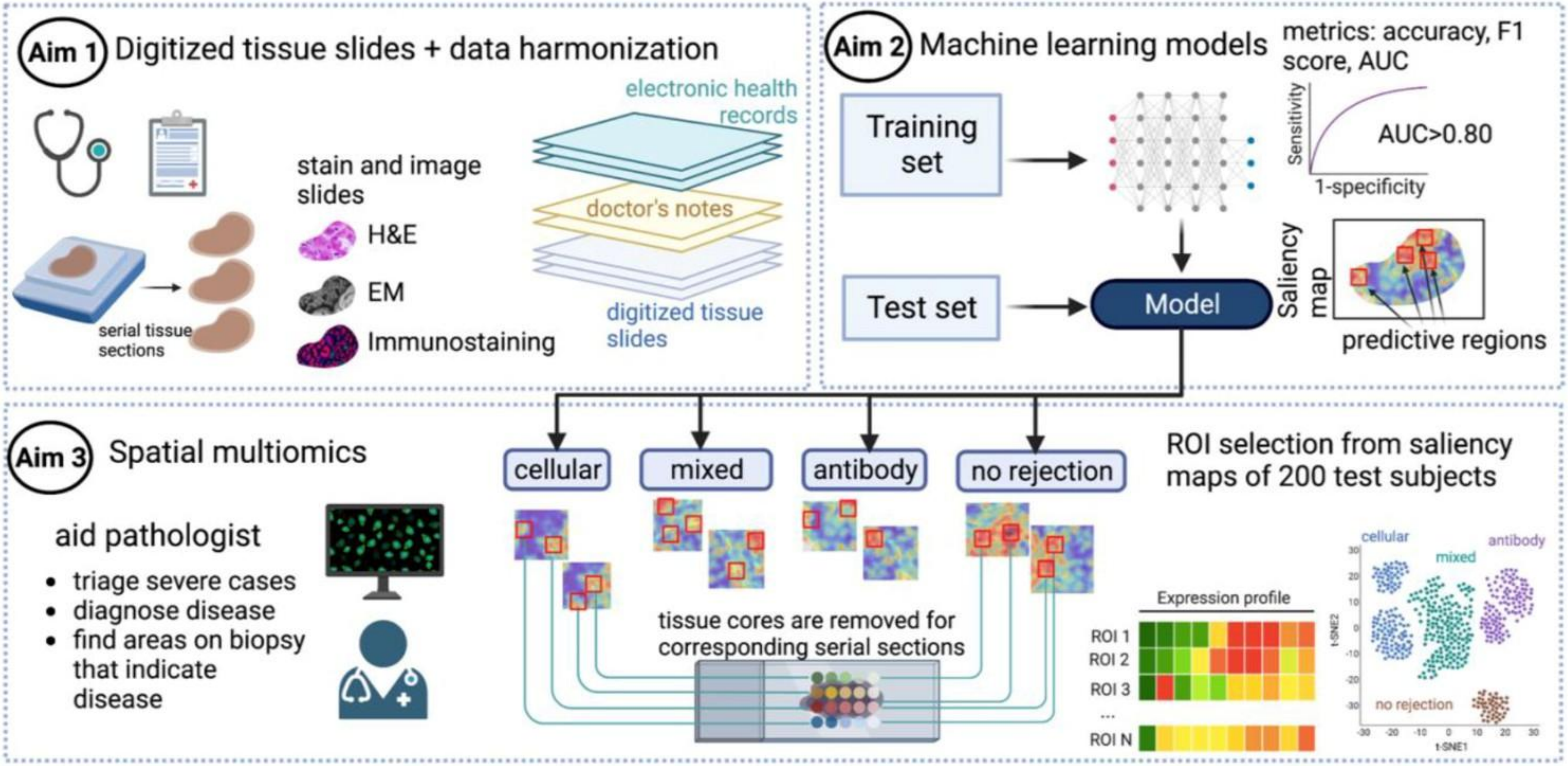


- Cohort of thousands of samples from hundreds of patients, majority from PROOF
- Composed of different types of no rejection, ACR, AMR, mixed rejection
- BMCB Currently retrieving slides from storage and digitizing them

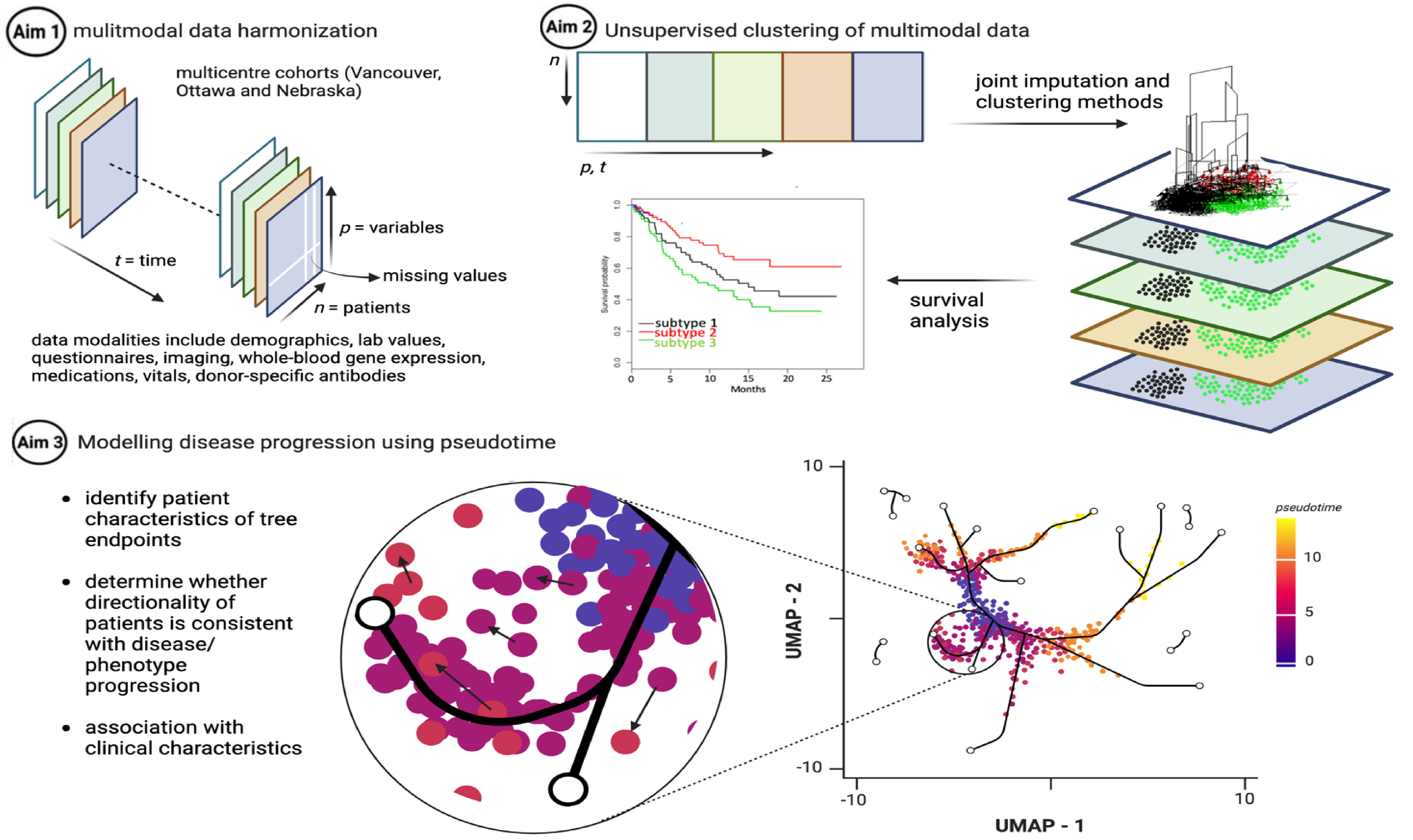
Data modalities

Variable-type	Variables
Demographics	Age, sex, ethnicity, comorbidities
Participant and donor viral serology	Positive/negative/not done: Epstein-Barr Virus (EPV), cytomegalovirus (CMV), Human Immunodeficiency Virus (HIV), Hepatitis C Virus (HCV), Hepatitis B Surface Antigen (HBsAg), Hepatitis B Surface Antibody (HBsAb), Hepatitis B Core Antigen (HBcAg), Hepatitis B Core Antibody (HBcAb), Hepatitis B Virus Deoxyribonucleic Acid (HBV DNA)
Induction therapy	Medication with dose, route and timelines
Vitals	Smoking status, weight, height, temperature, heart rate, systolic and diastolic blood pressure
Labs	Glucose, creatinine, estimated glomerular filtration rate, urea, sodium, potassium, chloride, bicarbonate, calcium, C reactive protein, Brain Natriuretic Peptide, N-Terminal pro-B-type Natriuretic Peptide, hemoglobin, platelet
Complete blood counts with differentials	Hemoglobin, platelets, white blood cells, red blood cells, neutrophils, hematocrit, lymphocytes, monocytes, eosinophils, basophils, mean corpuscular volume (MCV), red cell distribution width (RDW), mean platelet volume (MPV)
Immunosuppressant blood levels	Time and dose of tacrolimus, cyclosporine, sirolimus
Human Leukocyte Antigen – Panel Reactive Antibody (HLA-PRA)	Participant and donor HLA alleles: A1, A2, B, C, DR, DQ, DP
Medication	Medication times, dosing, route for both maintenance immunosuppression, hypertension and statin therapy
EMB biopsy	ISHLT grade, antibody mediated rejection, Quilty lesions, late ischemic injury
Graft outcomes	Echocardiographic data (ejection fraction, diastolic function (e.g. E/A, E/e’), chamber volumes (e.g. LV end-diastolic volume), cardiac allograft vasculopathy

Objective 1 – Improve diagnosis using multimodal data

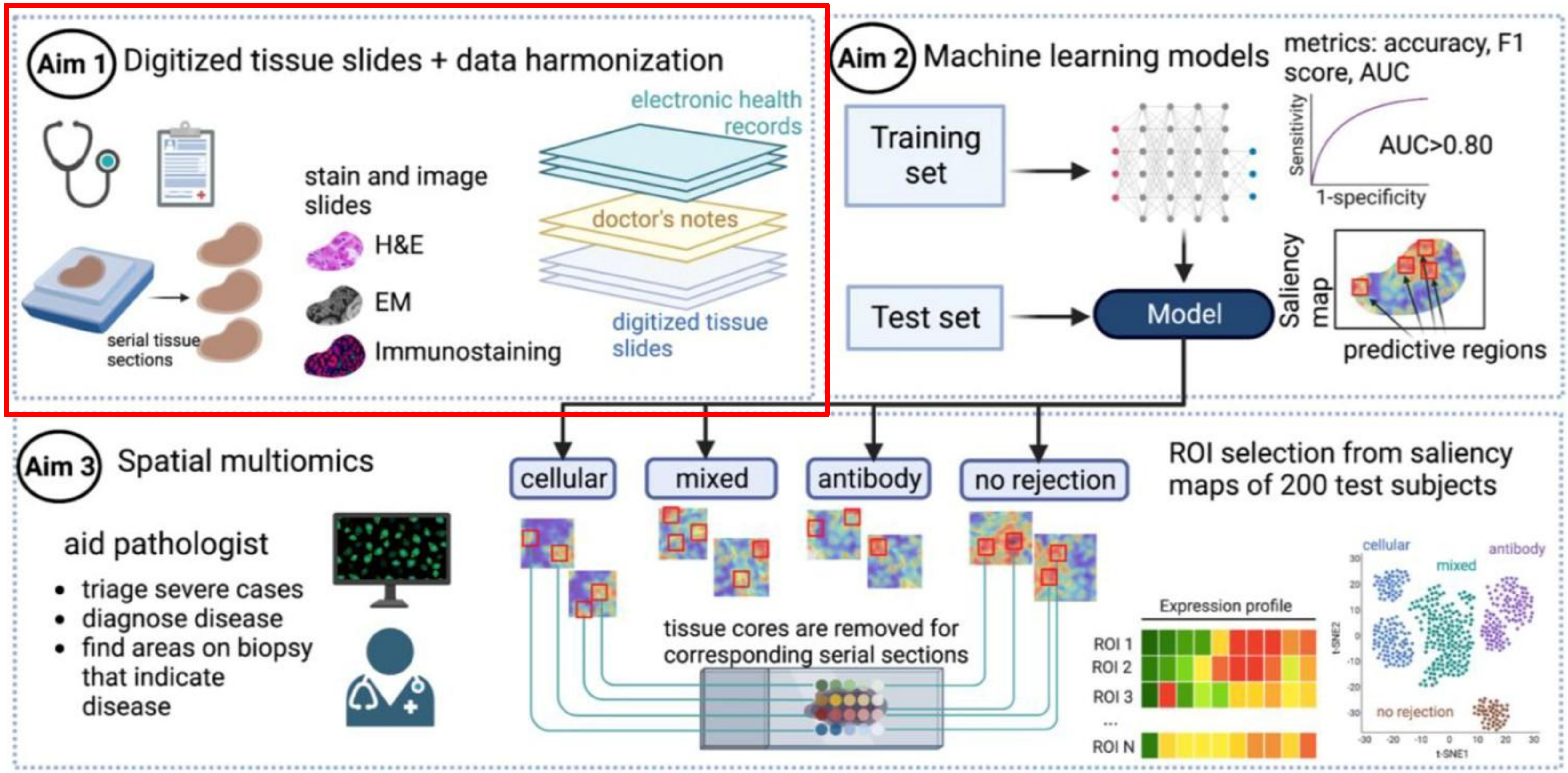


Objective 2 – Identify patient trajectories with distinct clinical outcomes



Progress To Date

Objective 1 – Improve diagnosis using multimodal data



Objective 1.1 – HTR Prediction using digital pathology

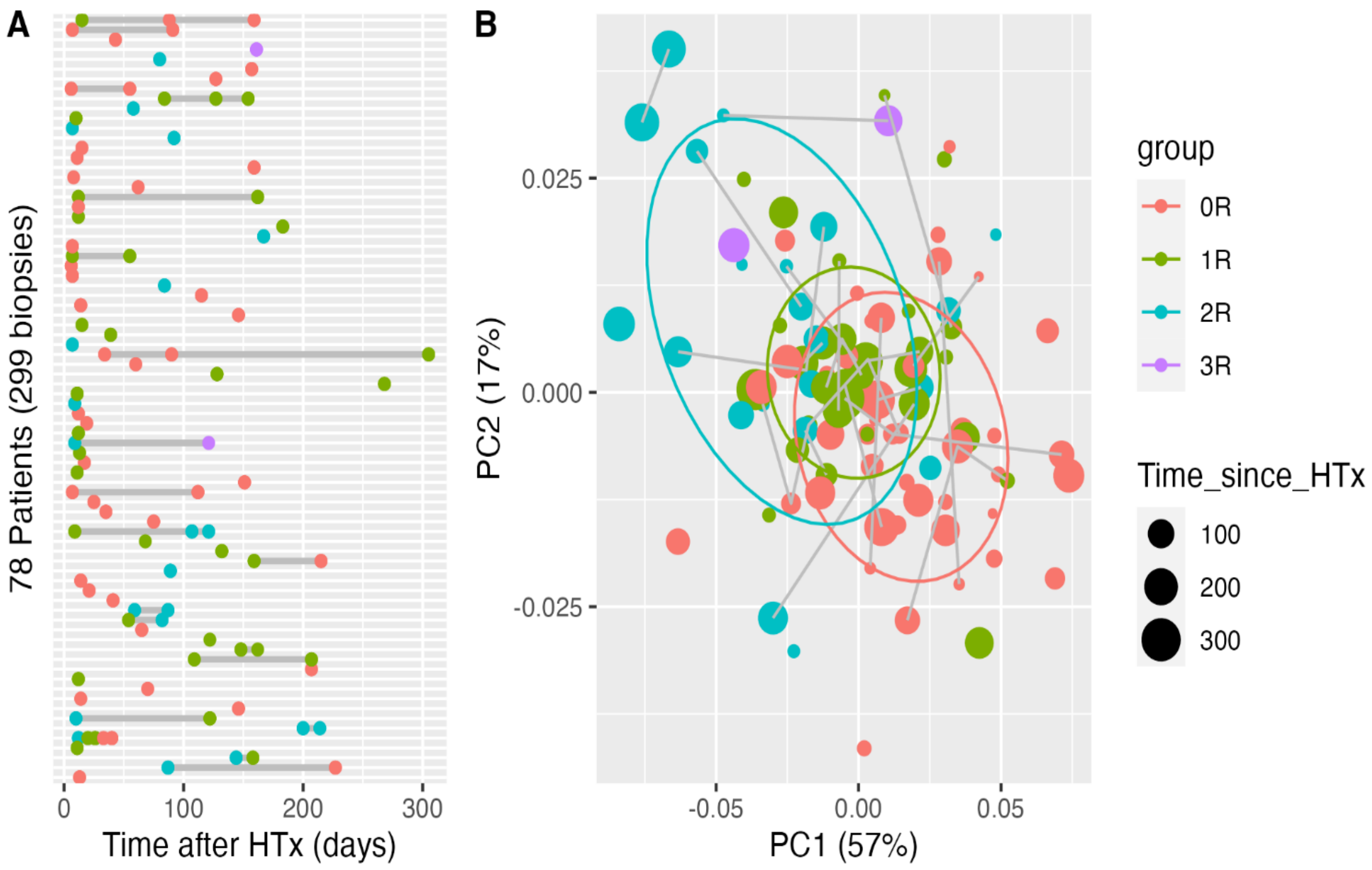
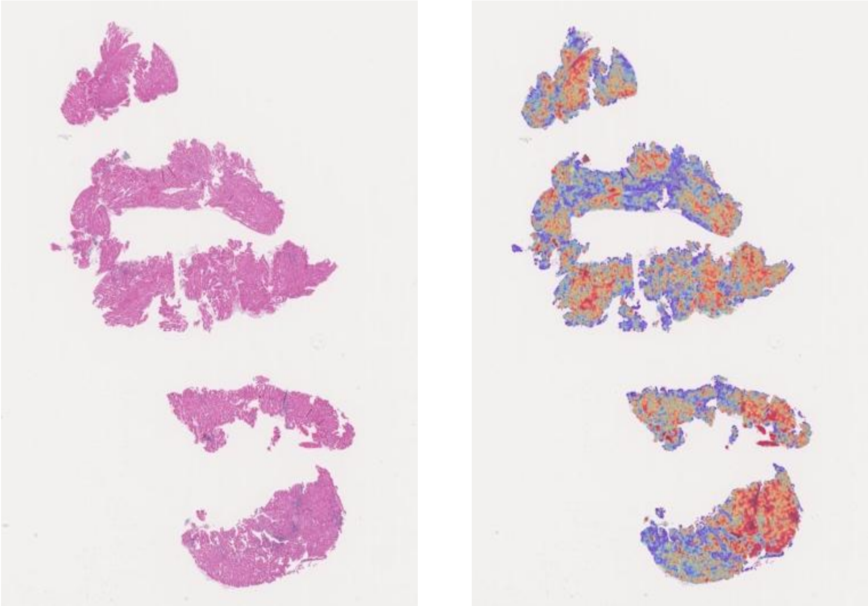


Image analysis of a subset of patients with heart transplant rejection. A. Patient biopsy timepoints post-transplant colored by ACR grade. B. Principal Component Analysis of significant features from H&E images depicts a separation with respect to ACR grade. ACR Gradings: 0R no rejection, 1R mild, 2R moderate, and 3R severe.

- Biopsies are more frequent early post-transplant (first 100 days)
- Clear separation between no rejection VS moderate rejection
  - But overlap exists, reflecting biological variability and diagnostic challenge

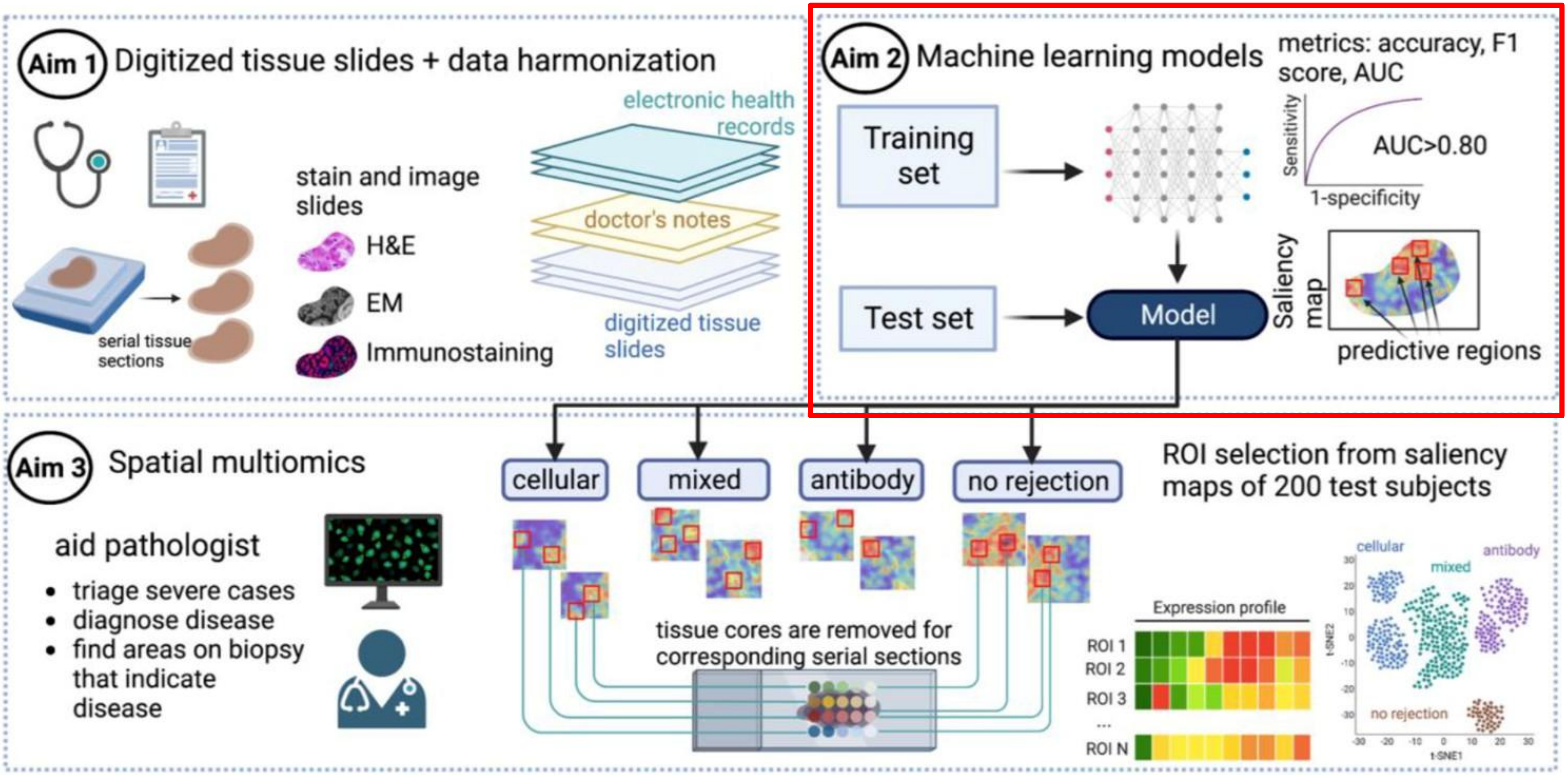
Sample saliency map



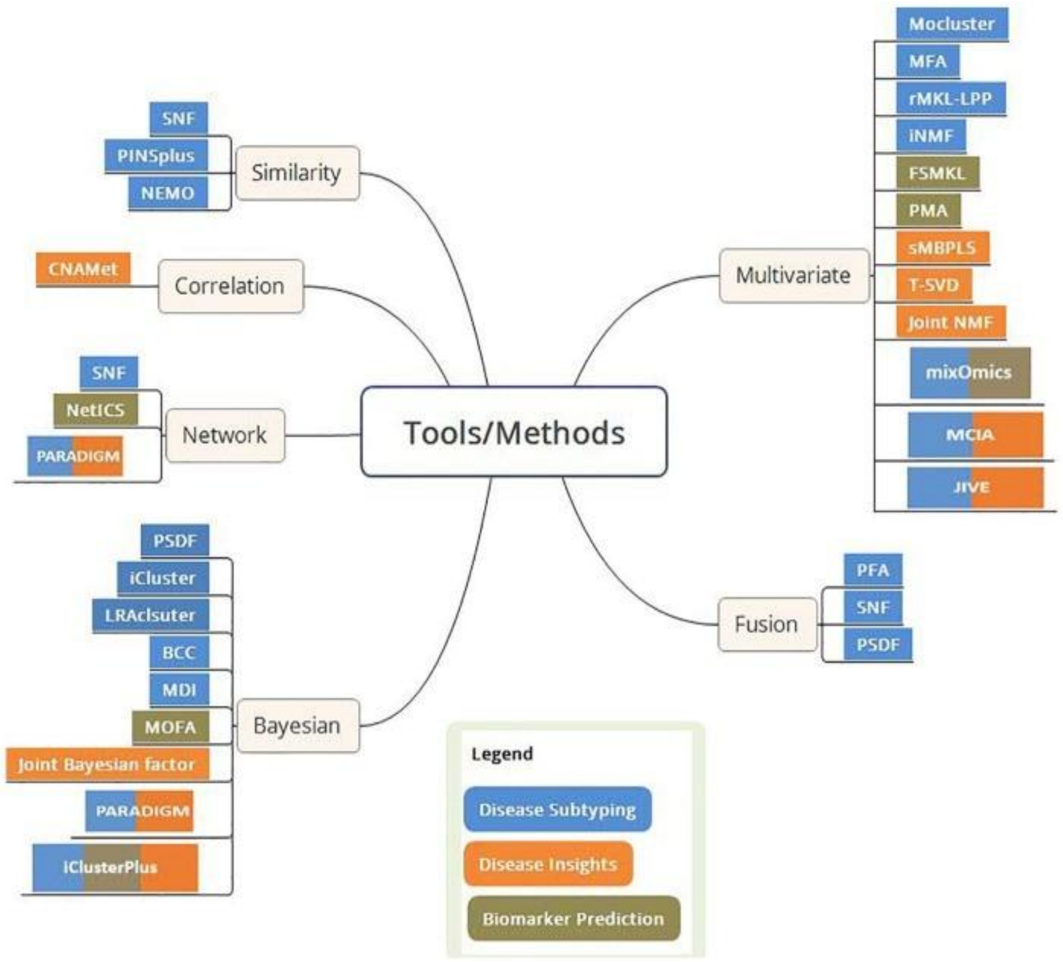
Model interpretation. Left - Original H&E slide of endomyocardial biopsy. Right - Saliency map of the H&E slide

- Saliency maps reveal WHERE the model detects pathology (explainable AI)
- AI model identifies cardiac transplant rejection from standard H&E biopsies
- Red areas = rejection zones; blue areas = normal tissue

Objective 1 – Improve diagnosis using multimodal data



Objective 1.2 – benchmarking multimodal models



Subramanian et al. Types of multimodal integration methods (9)

Question Arises

- Many integration methods

—> Which to use, how to choose them

- A reproducibility crisis

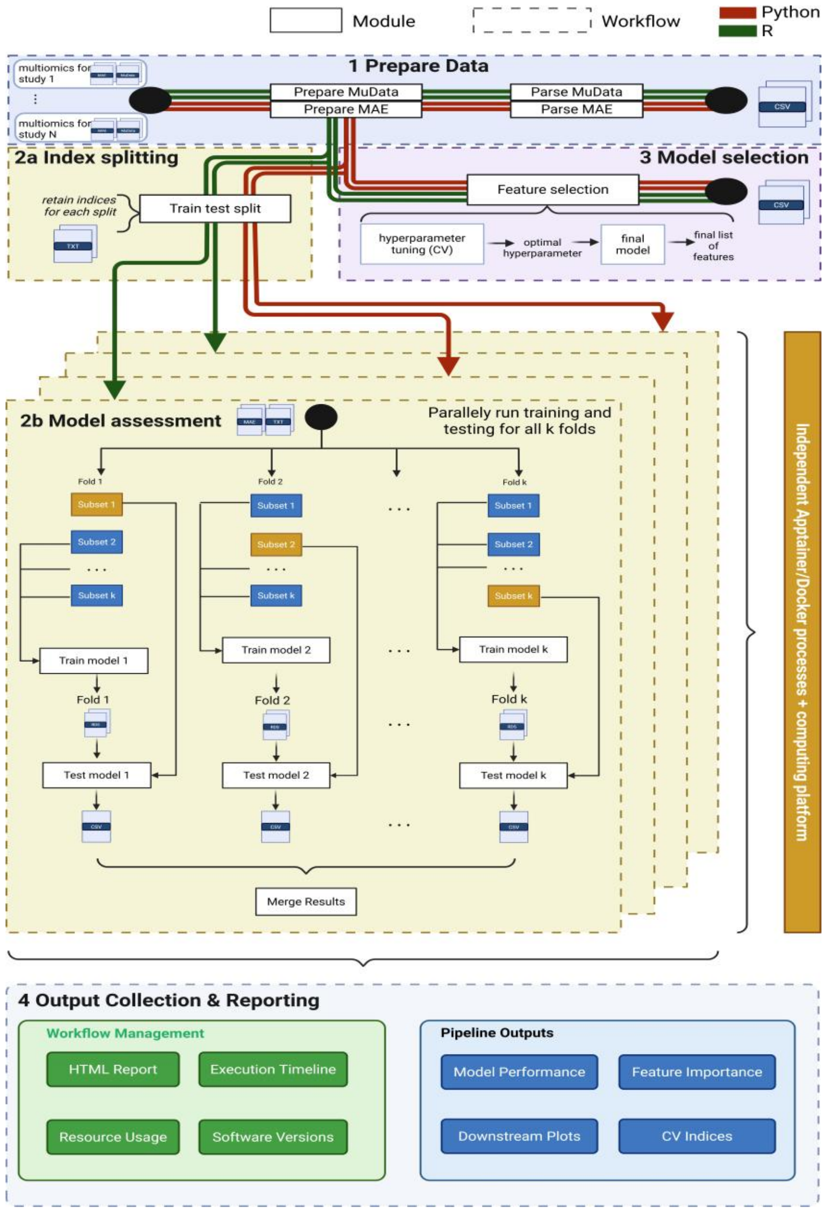
—> How to reproduce method and get reliable results?

- Existing benchmark studies are not 100% complete or all-encompassing

—> Technical difficulty in implementation?

Objective 1.2 – MESSI: benchmarking pipeline

Multimodal Experiments with SyStematic Interrogation using nextflow

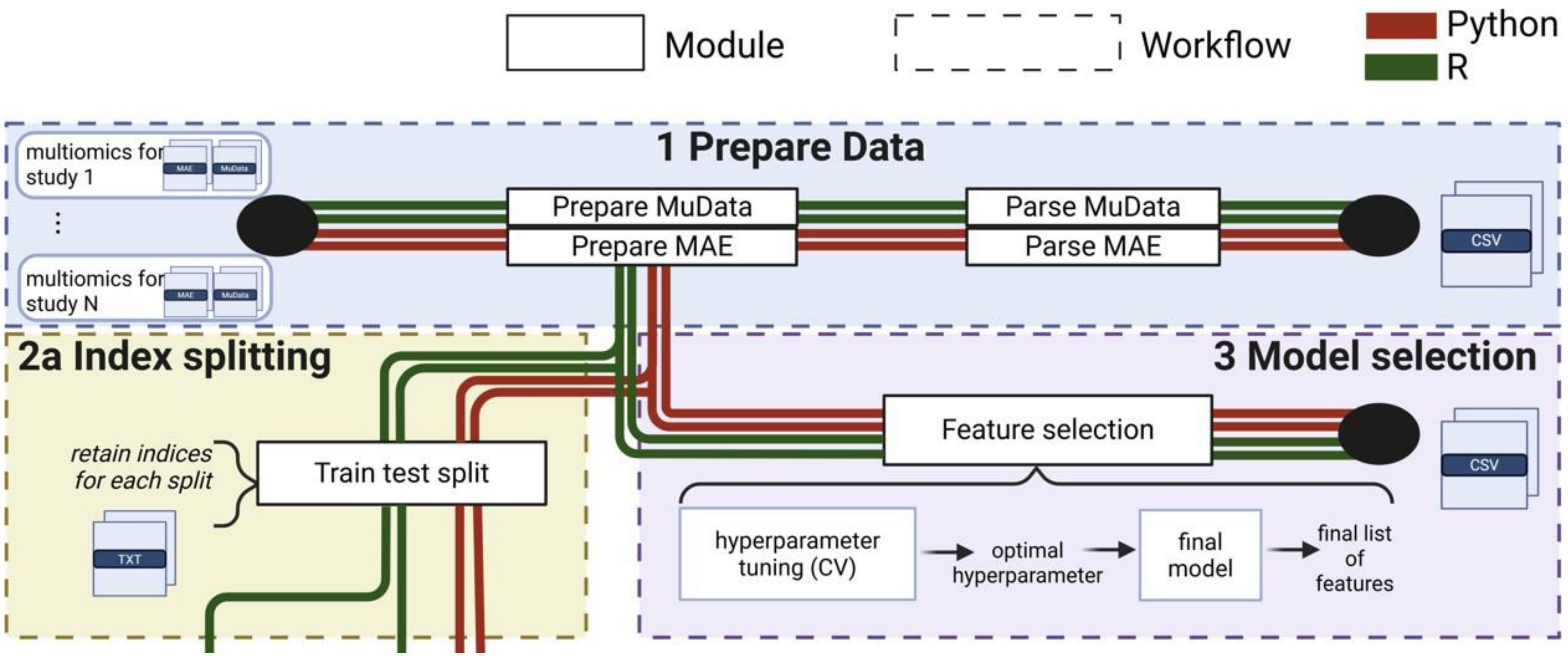


Stage 1

Stage 2

Stage 3

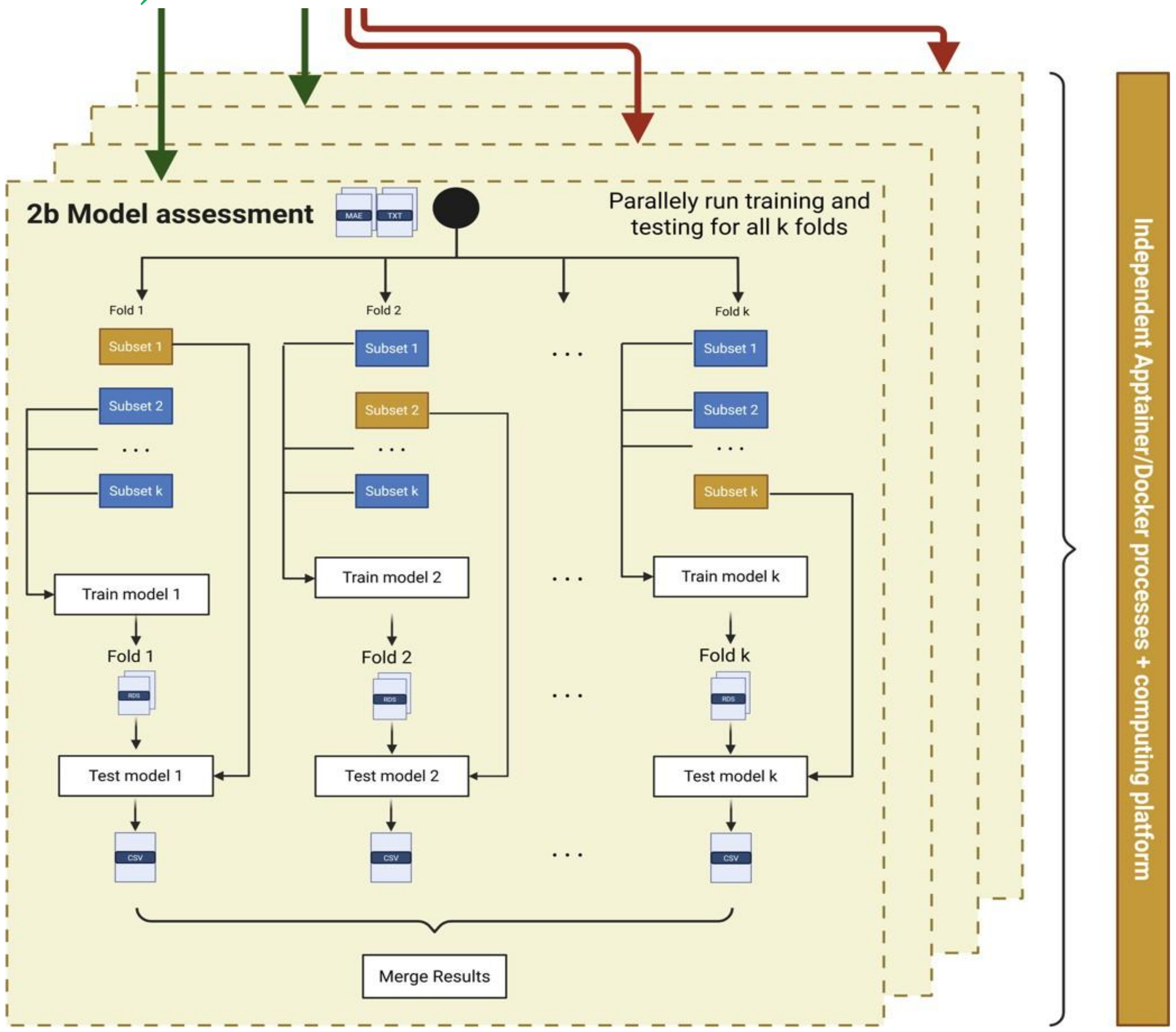
Objective 1.2 – MESSI: benchmarking pipeline



Stage 1

- Currently handles both R and Python –> could extend to more
- Data flows in there, handles N different datasets
- Important model selection prior to evaluation

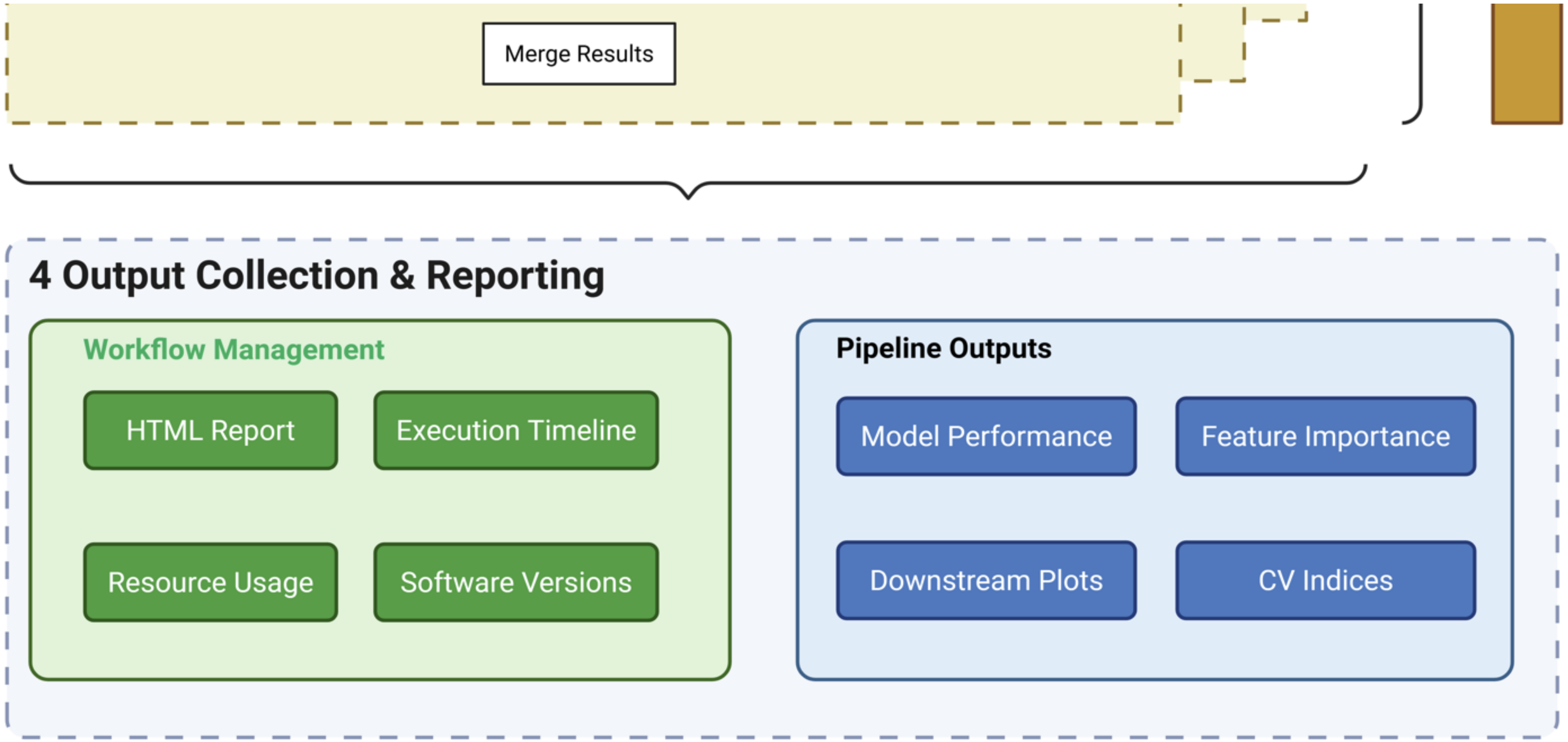
Objective 1.2 – MESSI: benchmarking pipeline



Stage 2

- Each method is an isolated workflow (dash box)
- Each method internally runs evaluation in parallel
- Easily extended to other tasks not just cross-validation

Objective 1.2 – MESSI: benchmarking pipeline



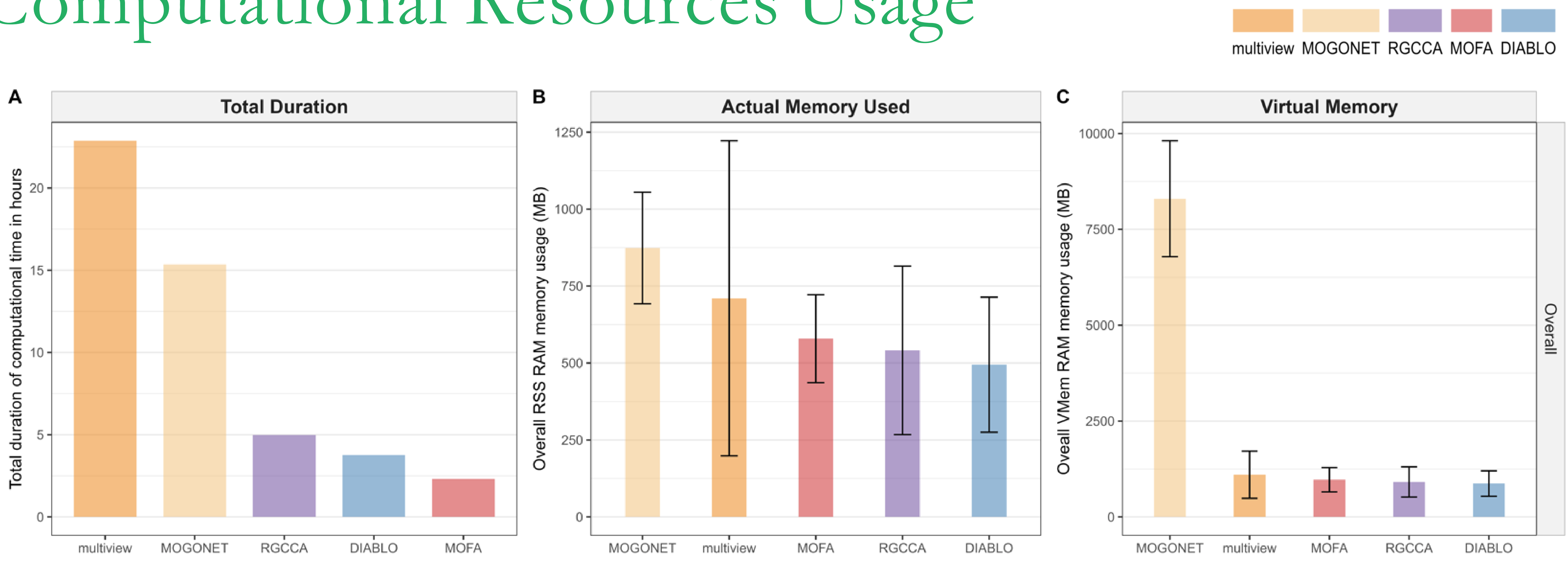
Stage 3

- “Recursively” collects output from each method including versions
- Summarizes metrics for downstream analysis
- Provides rich report of computational resources usages

Datasets evaluated

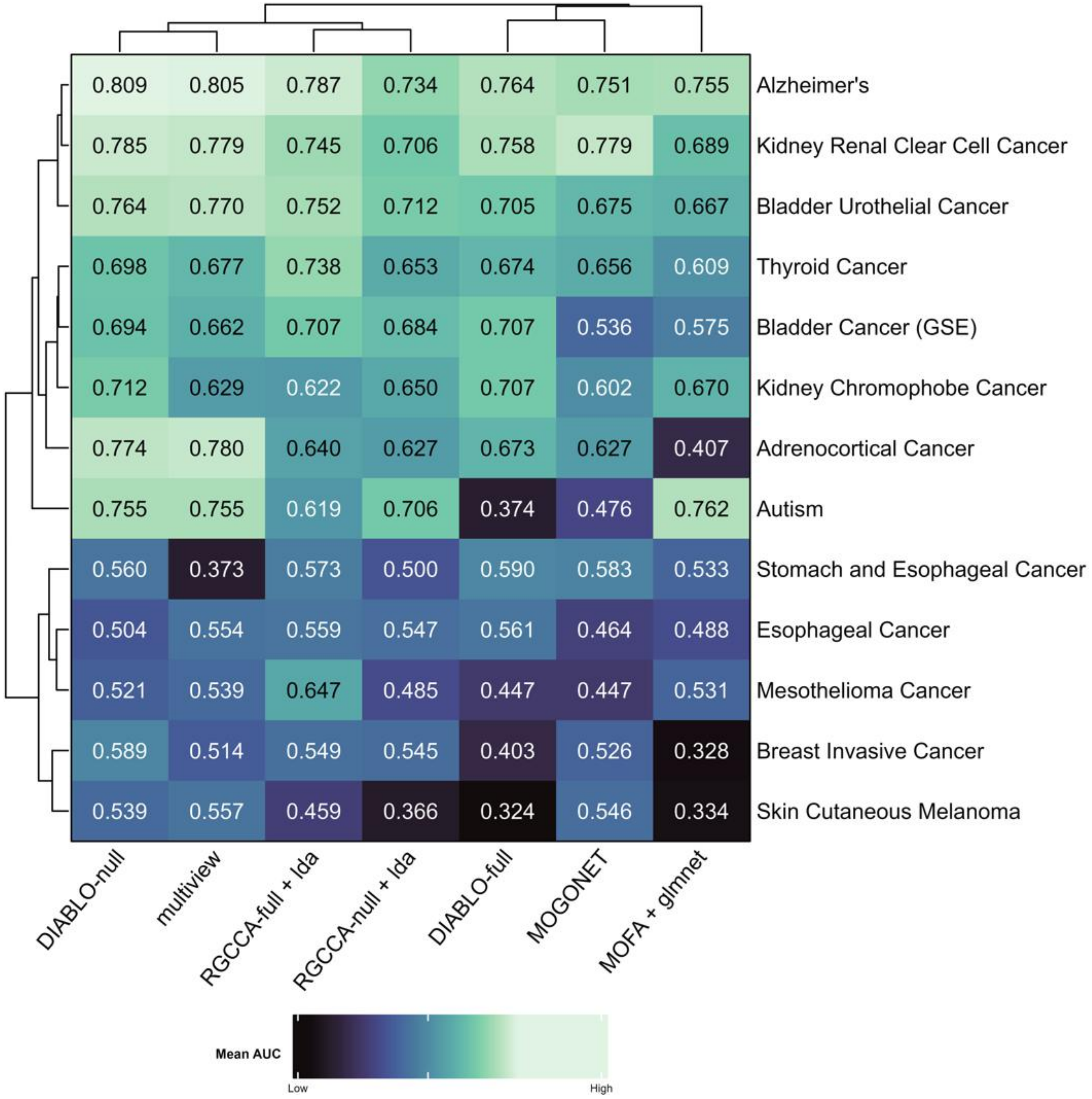
Disease	Number of Observations	Y=0	Y=1	Prop(Y=1)	Omics data (number of predictors)	Dataset
Autism	24	Control Cer	Autistic	0.458	mrna (8110), cpg (1674), cc (12)	GSE38609
Stomach and Esophageal Cancer	25	Stage I/II	Stage III/IV	0.600	cpg (4796), mirna (175), rppa (42), mrna (5590)	TCGA-STES
Bladder Cancer	33	Non-invasive bladder cancer	Invasive bladder cancer	0.424	mrna (5831), cpg (8915), cc (10)	GSE71669
Adrenocortical Cancer	46	Stage I/II	Stage III/IV	0.391	cpg (7396), mirna (166), rppa (40), mrna (5660)	TCGA-ACC
Adenomas and Adenocarcinomas	63	Stage I/II	Stage III/IV	0.302	cpg (6489), mirna (180), rppa (57), mrna (5517)	TCGA-KICH
Mesothelioma	63	Stage I/II	Stage III/IV	0.762	cpg (7647), mirna (203), rppa (41), mrna (5852)	TCGA-MESO
Skin Cutaneous Cancer	80	Stage I/II	Stage III/IV	0.350	cpg (7534), mirna (218), rppa (47), mrna (5691)	TCGA-SKCM
Breast Invasive Cancer	109	Stage I/II	Stage III/IV	0.358	cpg (7985), mirna (251), rppa (43), mrna (5936)	TCGA-BRCA
Esophageal Cancer	119	Stage I/II	Stage III/IV	0.353	cpg (7963), mirna (202), rppa (44), mrna (5709)	TCGA-ESCA
Kidney Cancer	123	Stage I/II	Stage III/IV	0.520	cpg (7726), mirna (214), rppa (44), mrna (5941)	TCGA-KIRC
Thyroid Cancer	217	Stage I/II	Stage III/IV	0.318	cpg (6367), mirna (231), rppa (25), mrna (5537)	TCGA-THCA
Bladder Cancer	336	Stage I/II	Stage III/IV	0.685	cpg (8072), mirna (268), rppa (44), mrna (6093)	TCGA-BLCA
Alzheimer's Disease	351	Normal control	Alzheimer's disease	0.519	cpg (58), mirna (74), mrna (90)	ROSMAP

Computational Resources Usage



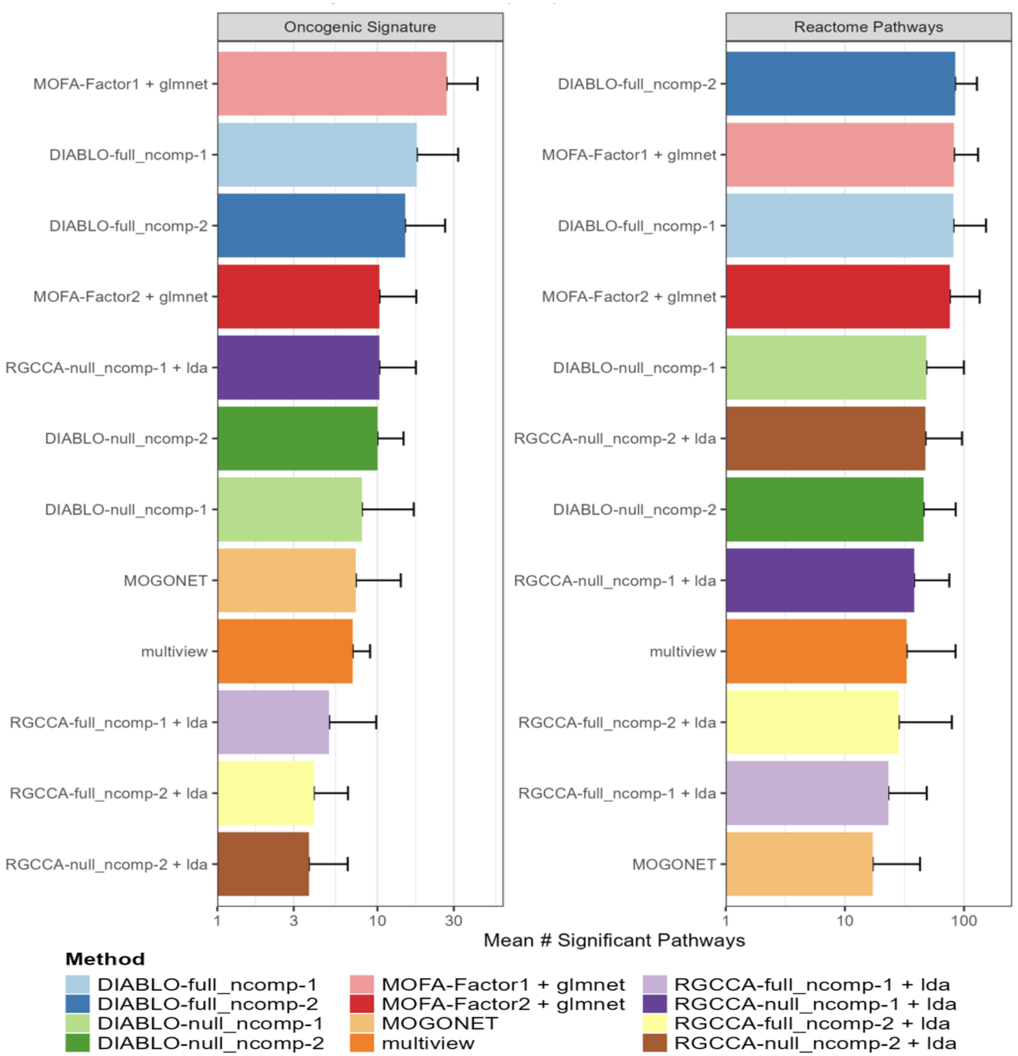
- MOGONET and multiview require longer durations compared to rest
- MOGONET consumes more virtual memory than actual memory due to its DL nature
- DIABLO takes less computational resources in general

Performance on real-world datasets



- “No method performs well in all scenarios”
- The DIABLO-null ranks top so far
- The MOFA + glmnet is weakest now

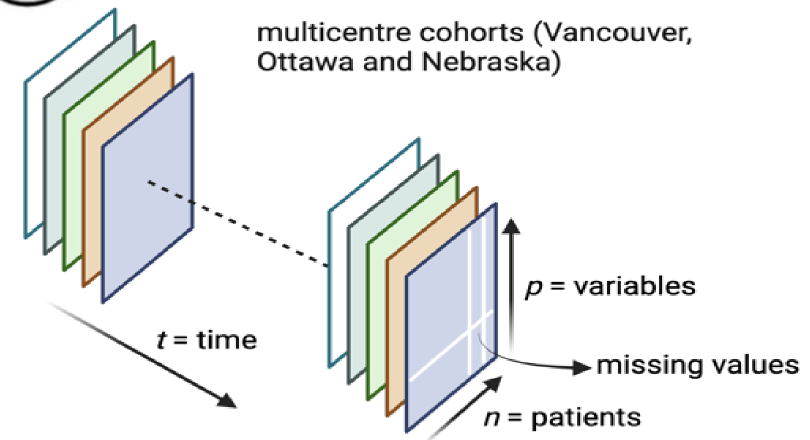
Biological Enrichment analysis



- DIABLO-full yields top in identify the highest number of significant pathways after GSEA with FDR < 0.2
- MOFA despite low predictive performance, ranks high in biological enrichment analysis
- Modelling assumption can affect pathway discovery sensitivity

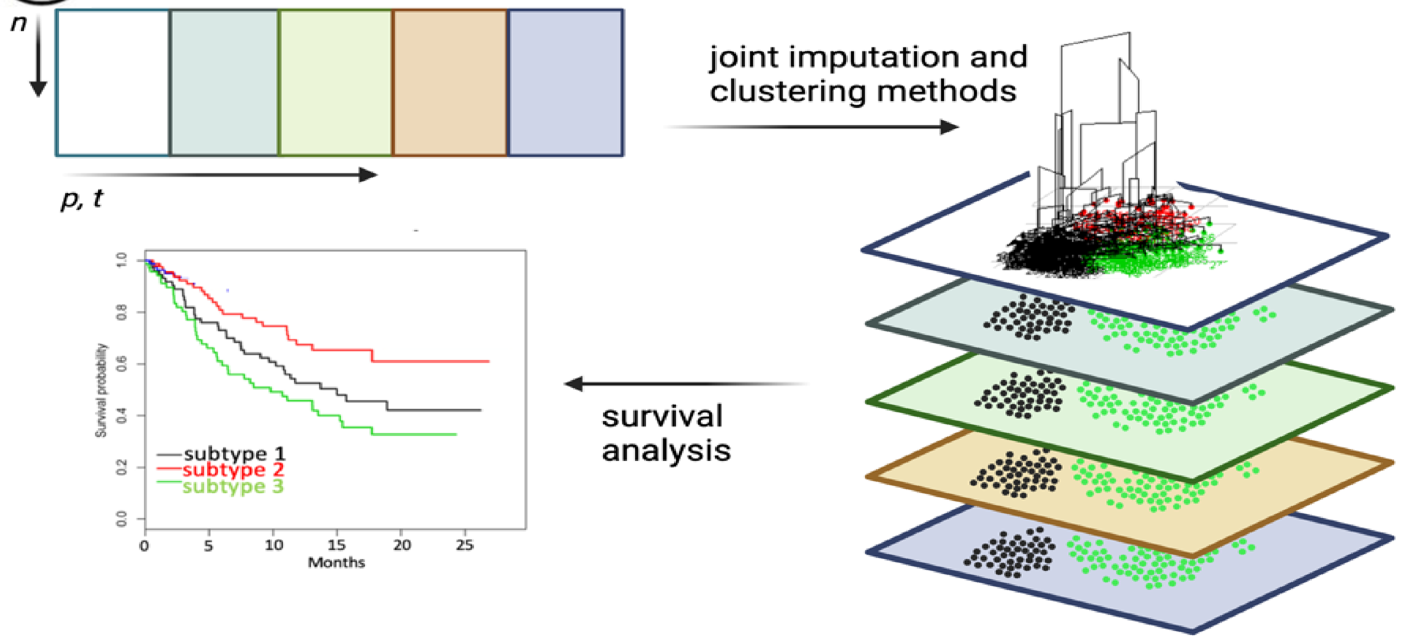
Objective 2 – Identify patient trajectories with distinct clinical outcomes

Aim 1 mulitmodal data harmonization



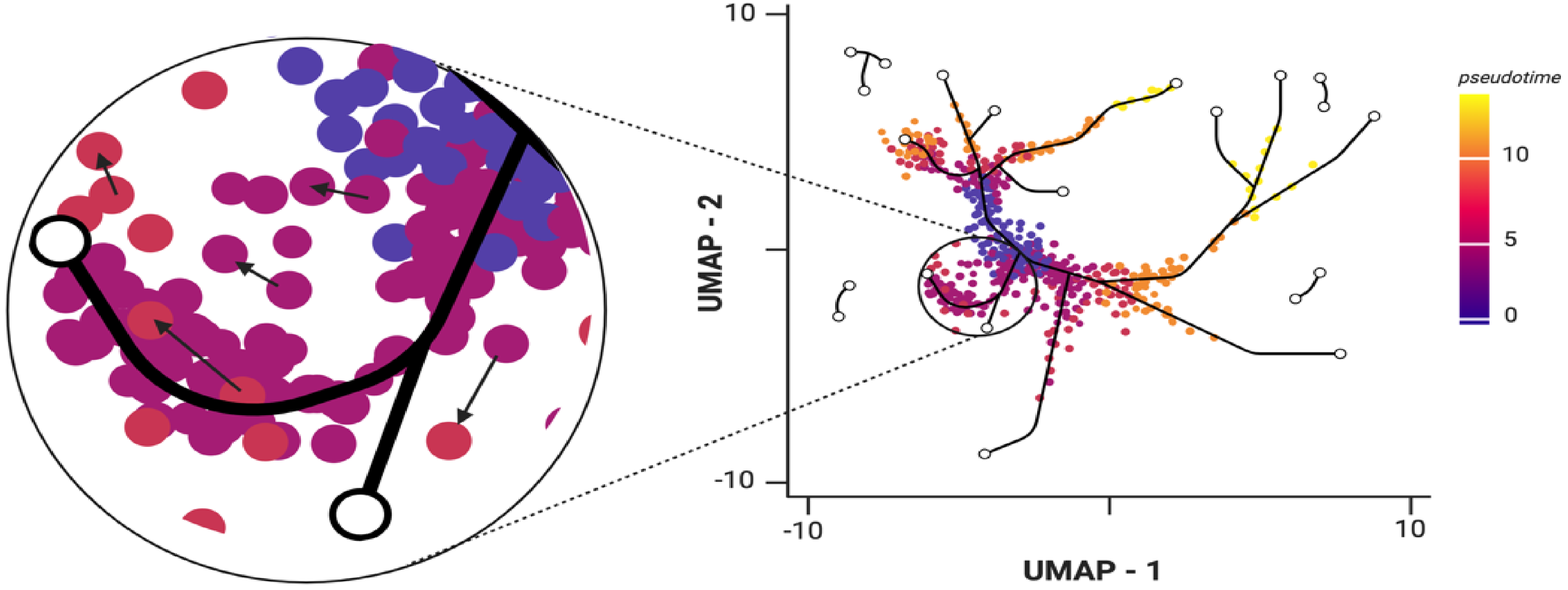
data modalities include demographics, lab values, questionnaires, imaging, whole-blood gene expression, medications, vitals, donor-specific antibodies

Aim 2 Unsupervised clustering of multimodal data

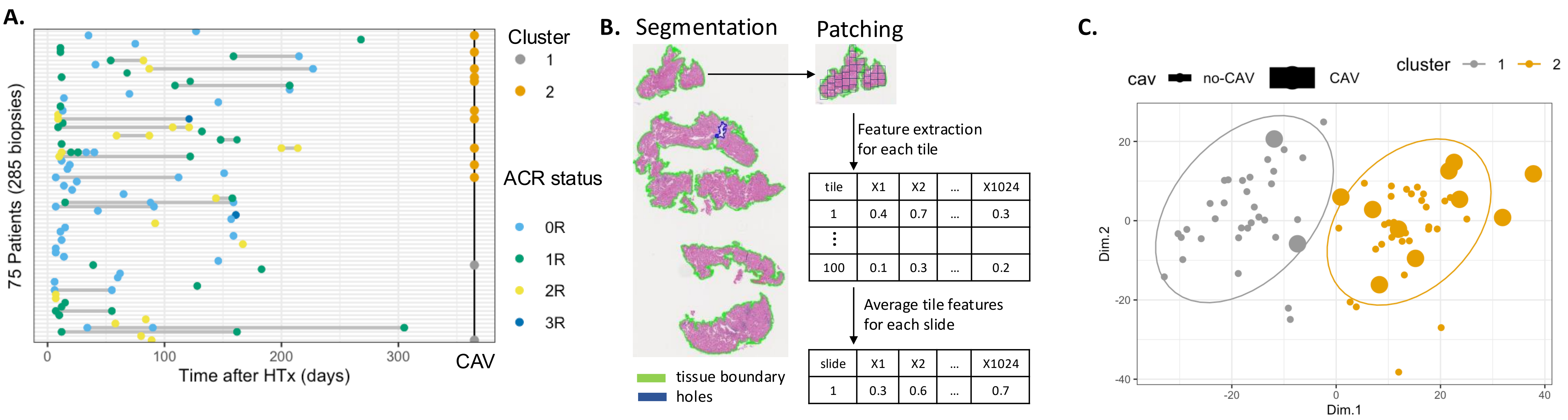


Aim 3 Modelling disease progression using pseudotime

- identify patient characteristics of tree endpoints
- determine whether directionality of patients is consistent with disease/ phenotype progression
- association with clinical characteristics



Objective 2 – preliminary analysis



- Recorded 285 biopsies from 75 patients after heart transplantation
- 12 / 75 patients developed CAV , and 10 / 12 CAV patients belonged to cluster 2 after PCA
- This suggests there are some group factors associated with prognosis of CAV

Conclusion

- Current HTR diagnosis is subjective with poor interobserver agreement (κ = 0.31-0.39)
- MESSI pipeline enables reproducible benchmarking of multimodal integration methods
- No universal "best" method, classical approaches often match deep learning performance
- Preliminary clustering identified distinct risk groups (10/12 CAV patients in cluster 2)
  - Suggests potential for prognostic stratification using multimodal data
- Explainable AI can assist pathologists in identifying rejection zones
- Multimodal integration shows promise for personalized risk assessment

Future Direction

Objective 1

- *Apply ensemble model to integration multimodal data for HTx prediction*
- *Do spatial transcription on predictive areas*

Objective 2

- *Integration of all modalities to and find patient clusters*
- *Associated them with clinical outcomes*

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- Dr. Scott Tebutt

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Reference

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